Module 3 — Quiz B (Concept Check)

Question bank · After Read 2 · open notes · unlimited attempts

Module 3, Quiz B: Concept check

Reading: Module 3 Read 2 (complete Read 1 + Quiz A first)
Canvas: Concept Check quiz (Module 3)

Apply graph selection, association description, and two-way table concepts from Read 2.

Full integration and demo-lab R patterns → Synthesis.

Section A — Graphs: identify and interpret (Read 2 §2–3 + Read 1)

A1. Which graph for numerical by group?

You have a **numerical** outcome (`body_mass_g`) measured for three species (categorical). Which graph best compares the distributions?

A2. Which graph for two numerical?

You have two **numerical** variables: `bill_length_mm` (x) and `flipper_length_mm` (y). Which graph?

A3. Which graph for two categorical?

You have two **categorical** variables: `species` and `island`. Which graph shows how they co-occur?

A4. Read a boxplot

A boxplot has lower fence 10, $Q_1 = 18$, median $= 25$, $Q_3 = 32$, upper fence 45. What is the **IQR**?

A5. Skew from boxplot

A boxplot shows: min=5, $Q_1 = 10$, median=12, $Q_3 = 20$, max=50. Is the distribution skewed? Which way?

Answer key — Section A — Graphs: identify and interpret (Read 2 §2–3 + Read 1)

1. **Boxplot per group** (side-by-side boxplots with `group` on x-axis) — or grouped histograms/dot plots. Boxplots clearly show median, spread, and outliers per group.
2. A **scatterplot** (`geom_point()` with `aes(x = bill_length_mm, y = flipper_length_mm)`).
3. A **stacked bar chart** (or 100% stacked bar) using `geom_col()` with `fill = species` grouped by `island`.
4. IQR $= Q_3 - Q_1 = 32 - 18 = 14$.
5. **Right-skewed (positively skewed)** — the upper whisker (12 to 50) is much longer than the lower whisker, and the median is closer to Q_1 than Q_3.

Section B — Choosing the right summary by variable type

B1. Two numerical variables

Both variables are **numerical** (`height` and `weight`). What is the first graph to make? What do you describe?

B2. Two categorical variables

Both variables are **categorical** (`diet_type` with 3 levels, `outcome` with 2 levels). Best graph and summary?

B3. Numerical + categorical

One variable is **numerical** (`blood_pressure`) and one is **categorical** (`treatment_group`: A/B/C). Best graph?

B4. Univariate: one categorical

One variable: `habitat` with levels Urban/Suburban/Rural. Best graph and summary?

Answer key — Section B — Choosing the right summary by variable type

1. **Scatterplot.** Describe **form** (linear/curved/none), **direction** (positive/negative/none), **strength** (strong/moderate/weak).
2. **Stacked bar chart** (counts or 100% proportions). **Two-way table** of counts and row/column proportions.
3. **Side-by-side boxplots** (`geom_boxplot()` with group on x-axis) or grouped histograms. Summarize with `group_by(treatment_group) %>% summarize()`.
4. **Bar chart** (`geom_bar()` or `count()` + `geom_col()`). Summary: **counts and proportions** per level.

Section C — Association description and two-way tables (Read 2)

C1. Positive association

Describe a **positive** association between two numerical variables in words.

C2. Form vs direction

What is the difference between **form** and **direction** in a scatterplot description?

C3. Row proportion

Two-way table: Robin Urban=8, Robin Rural=2 (row total=10). Sparrow Urban=4, Sparrow Rural=6 (row total=10). What is the **row proportion** of Sparrows that are Rural?

C4. Column proportion

Two-way count table (species × habitat): Robin Urban=8, Robin Rural=2. Sparrow Urban=4, Sparrow Rural=6. Column totals: Urban=12, Rural=8. What is the **column proportion** of Urban birds that are Robin?

C5. Joint vs marginal

What is the difference between a **joint** proportion and a **marginal** proportion in a two-way table?

C6. 100% stacked bar

When would you prefer a **100% stacked bar** (`position = 'fill'`) over a regular stacked bar?

Answer key — Section C — Association description and two-way tables (Read 2)

1. As one variable **increases**, the other tends to **increase** as well (upward trend in the scatterplot).
2. **Form** is the shape of the trend (linear, curved, no pattern). **Direction** is whether higher x goes with higher y (positive), lower y (negative), or no consistent change (none).
3. $6/10 = 0.60$ — 60% of Sparrows are in Rural habitat (denominator is Sparrow row total).
4. $8/12 \approx 0.667$ — about 67% of Urban birds are Robins (denominator is Urban column total).
5. **Joint:** fraction of ALL observations in one specific cell (divide by grand total). **Marginal:** fraction in a row or column total (divide by grand total for that row or column).
6. When you want to compare **proportions within groups** rather than raw counts — especially when group sizes differ, so the bars all reach 100% and the composition is easier to compare.
